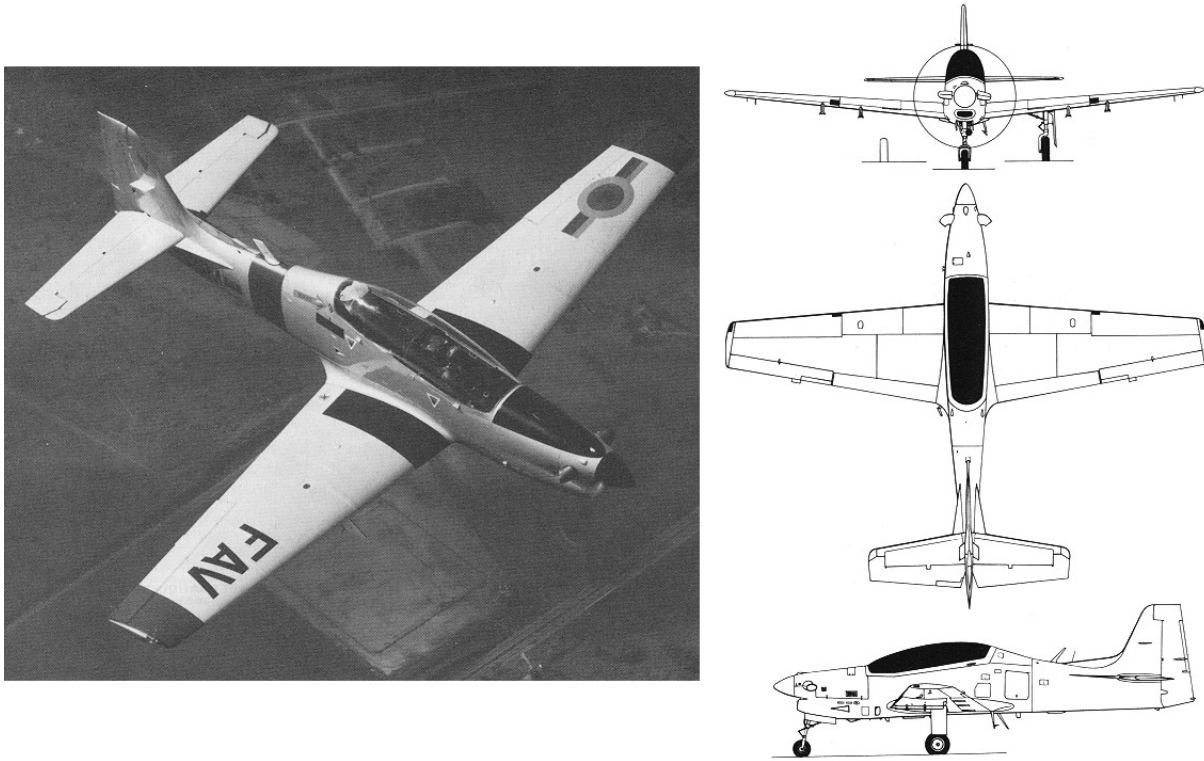




**Figure 9.4.1.** The tapered wing used on the Embraer EMB-312 Tucano turboprop. (Photograph by Barry Santana.)

#### 9.4. Tapered Wings

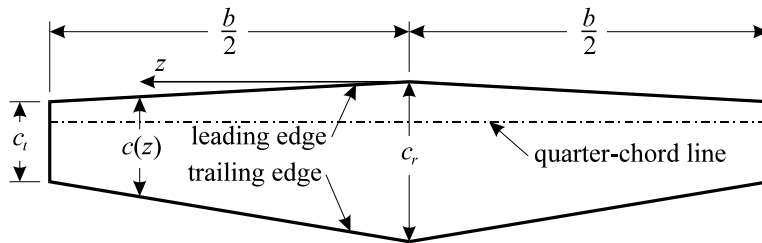
While elliptic wings produce the minimum possible induced drag, they are more expensive to manufacture than simple rectangular wings. Rectangular wings, on the other hand, are easy to manufacture, but they generate induced drag at a level that is less than optimum. The tapered wing, shown in Fig. 9.4.1, is a compromise. Tapered wings have a chord length that varies linearly from the root to the tip. They are nearly as easy to manufacture as rectangular wings, and they can be designed to produce induced drag close to the optimum value of an elliptic wing.

Consider a tapered wing with no geometric twist and no aerodynamic twist, as shown in Fig. 9.4.2. The chord length varies with  $z$  according to the relation

$$c(z) = c_r - (c_r - c_t)|2z/b| = c_r[1 - (1 - R_T)|2z/b|] \quad (9.4.1)$$

where  $c_r$  is the root chord,  $c_t$  is the tip chord, and  $R_T$  is the taper ratio,

$$R_T = c_t/c_r \quad (9.4.2)$$



**Figure 9.4.2.** The planform shape of an unswept tapered wing with a taper ratio of 0.5 and an aspect ratio of 6.

The planform area for this wing is

$$S_p = \int_{z=-b/2}^{b/2} c(z) dz = 2c_r \int_{z=0}^{b/2} \left[ 1 - (1 - R_T) \frac{2z}{b} \right] dz = \frac{bc_r}{2} (1 + R_T) \quad (9.4.3)$$

Solving Eq. (9.4.3) for the root chord length, we have

$$c_r = \frac{2S_p}{b(1 + R_T)} = \frac{2b}{R_A(1 + R_T)} \quad (9.4.4)$$

Using Eq. (9.4.4) in Eq. (9.4.1) gives

$$c(z) = \frac{2b}{R_A(1 + R_T)} \left[ 1 - (1 - R_T) \left| \frac{2z}{b} \right| \right] \quad (9.4.5)$$

Now, using the change of variables,

$$z = -\frac{1}{2}b \cos(\theta) \quad (9.4.6)$$

Eq. (9.4.5) can be written

$$c(\theta) = \frac{2b}{R_A(1 + R_T)} [1 - (1 - R_T) |\cos(\theta)|] \quad (9.4.7)$$

Using Eq. (9.4.7) in Eq. (9.2.9), we have

$$\sum_{n=1}^N B_n \sin(n\theta) \left[ \frac{2R_A(1 + R_T)}{\tilde{C}_{L,\alpha} [1 - (1 - R_T) |\cos(\theta)|]} + \frac{n}{\sin(\theta)} \right] = \alpha - \alpha_{L0} \quad (9.4.8)$$

Because we are considering a wing with no geometric or aerodynamic twist, the right-hand side of Eq. (9.4.8) is independent of  $\theta$ . Dividing Eq. (9.4.8) through by  $\alpha - \alpha_{L0}$  gives,

$$\sum_{n=1}^N b_n \sin(n\theta) \left[ \frac{2R_A(1 + R_T)}{\tilde{C}_{L,\alpha} [1 - (1 - R_T) |\cos(\theta)|]} + \frac{n}{\sin(\theta)} \right] = 1 \quad (9.4.9)$$

where

$$b_n = \frac{B_n}{\alpha - \alpha_{L0}} \quad (9.4.10)$$

The advantage of Eq. (9.4.9) over Eq. (9.4.8) is that we can solve for the values of the coefficients  $b_n$  independent of the angle of attack. This does not mean that the lift coefficient is independent of angle of attack. Using Eq. (9.4.10) in Eq. (9.2.3), we can write the circulation distribution as

$$\Gamma(\theta) = 2bV_\infty \sum_{n=1}^N B_n \sin(n\theta) = 2bV_\infty (\alpha - \alpha_{L0}) \sum_{n=1}^N b_n \sin(n\theta) \quad (9.4.11)$$

Using Eq. (9.4.10) with Eq. (9.2.14), we can write the lift coefficient as

$$C_L = \pi R_A B_1 = \pi R_A b_1 (\alpha - \alpha_{L0}) \quad (9.4.12)$$

It is convenient to rewrite Eq. (9.4.12) as,

$$C_L = C_{L,\alpha} (\alpha - \alpha_{L0}) \quad (9.4.13)$$

where

$$C_{L,\alpha} = \frac{\tilde{C}_{L,\alpha}}{(1 + \tilde{C}_{L,\alpha} / \pi R_A)(1 + \kappa_L)} \quad (9.4.14)$$

and

$$\kappa_L = \frac{1 - (1 + \pi R_A / \tilde{C}_{L,\alpha}) b_1}{(1 + \pi R_A / \tilde{C}_{L,\alpha}) b_1} \quad (9.4.15)$$

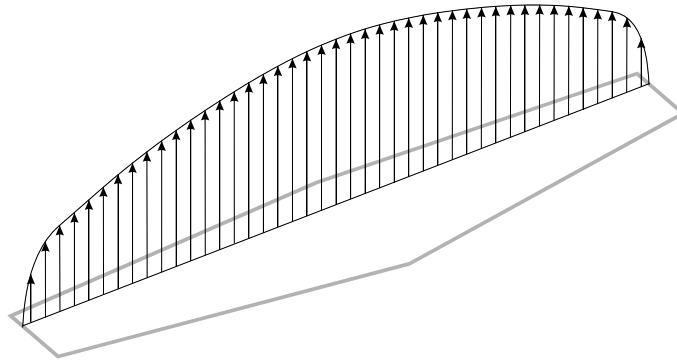
Using Eq. (9.4.10) with Eq. (9.2.19), we can write the induced drag coefficient as

$$C_{D_i} = \frac{C_L^2}{\pi R_A e_s} = \frac{C_L^2}{\pi R_A} (1 + \kappa_D) \quad (9.4.16)$$

where

$$\kappa_D = \sum_{n=2}^N n \left( \frac{b_n}{b_1} \right)^2 \quad (9.4.17)$$

Figure 9.4.3 shows the local section lift distribution for an unswept tapered wing with a taper ratio of 0.5 and an aspect ratio of 6. Numerical results for the variation in lift slope factor,  $\kappa_L$ , and induced drag factor,  $\kappa_D$ , with taper ratio and aspect ratio are plotted in Figs. 9.4.4 and 9.4.5 for a section lift slope of  $2\pi$ . Similar results were first published by the well-known English aerodynamicist, Hermann Glauert (1926). Notice from Fig. 9.4.5 that a tapered wing can be designed to produce induced drag at very near the minimum level of an elliptic wing. For typical aspect ratios, minimum induced drag is obtained at a taper ratio very near 0.35. At this taper ratio, the induced drag produced by a tapered wing is less than four percent greater than that produced by an elliptic wing and the lift slope for the tapered wing is diminished by less than one percent over that of the elliptic wing. Even with no taper at all, a rectangular wing of aspect ratio less than 12 increases the induced drag by less than 10 percent and decreases the lift slope by less than 4 percent when compared to an elliptic wing of the same aspect ratio.



**Figure 9.4.3.** The local section lift distribution for an unswept tapered wing with a taper ratio of 0.5 and an aspect ratio of 6.

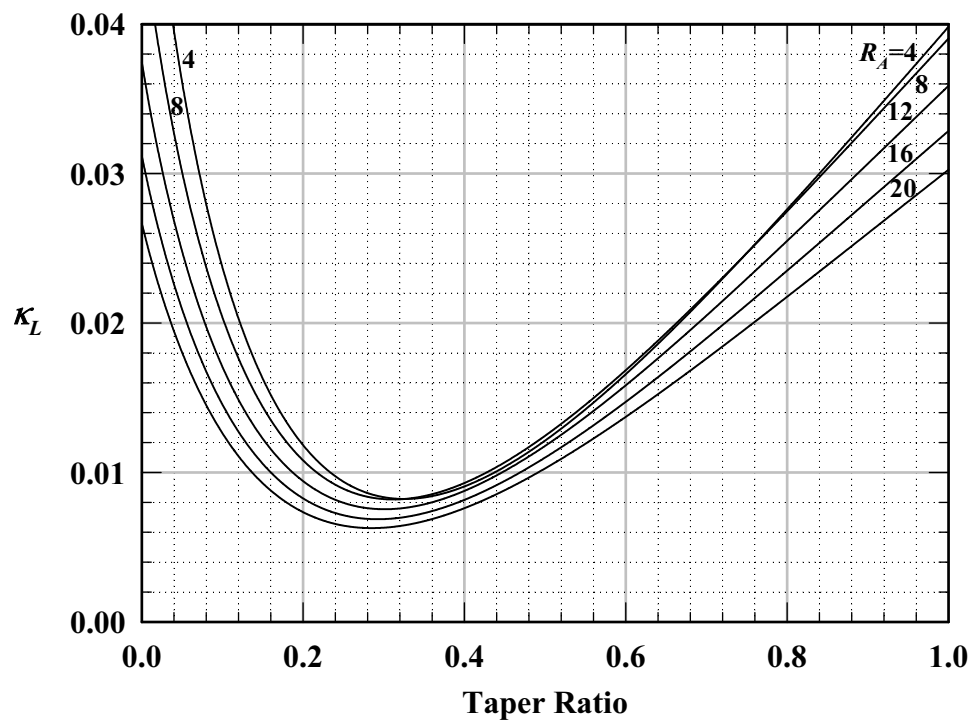


Figure 9.4.4. The lift slope factor from Prandtl's lifting-line theory.

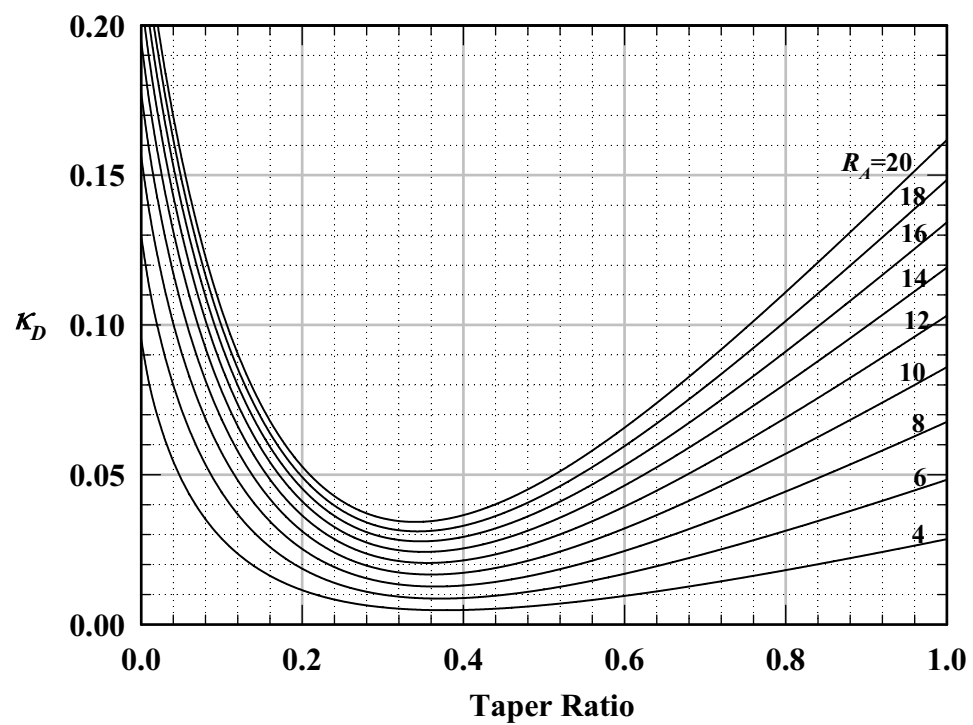
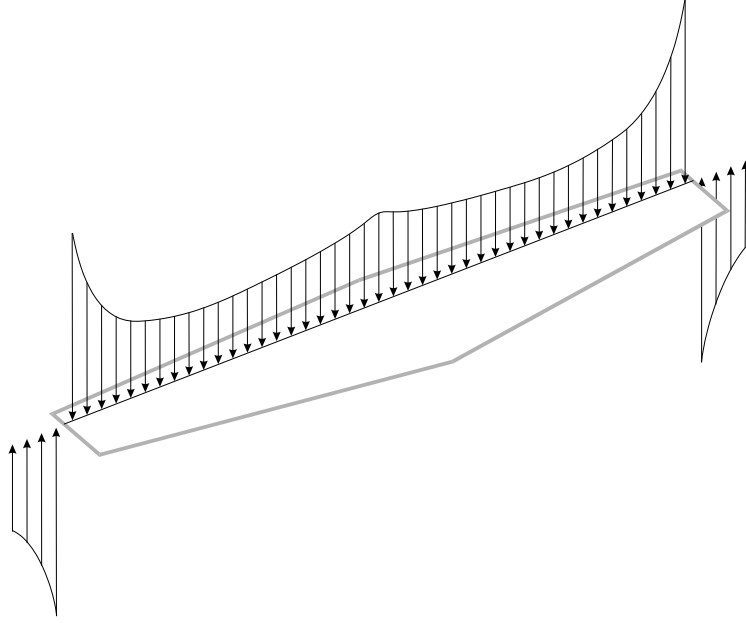


Figure 9.4.5. The induced drag factor from Prandtl's lifting-line theory.



**Figure 9.4.6.** The downwash along the axis of an unswept tapered wing with taper ratio of 0.5 and aspect ratio of 6.

The downwash induced near a tapered wing is found by using Eq. (9.4.10) in Eq. (9.2.26),

$$V_y(\bar{x}, \bar{y}, \bar{z}) = -\frac{V_\infty C_L}{\pi R_A} \sum_{n=1}^N \frac{b_n}{b_1} \frac{1}{\pi} \int_{\phi=0}^{\pi} \left\{ \frac{n \cos(n\phi) [\bar{z} + \cos(\phi)]}{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2} + \frac{n \bar{x} \cos(n\phi) [\bar{z} + \cos(\phi)]}{\{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2\} \{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{1/2}} + \frac{\bar{x} \sin(n\phi) \sin(\phi)}{\{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{3/2}} \right\} d\phi \quad (9.4.18)$$

The downwash distribution along the span of a tapered wing is,

$$V_y(0, 0, \bar{z}) = -\frac{V_\infty C_L}{\pi R_A} \sum_{n=1}^N \frac{b_n}{b_1} \frac{1}{\pi} \int_{\phi=0}^{\pi} \frac{n \cos(n\phi)}{\bar{z} + \cos(\phi)} d\phi = -\frac{V_\infty C_L}{\pi R_A} \sum_{n=1}^N \frac{b_n}{b_1} \frac{n \sin[n \cos^{-1}(-\bar{z})]}{\sin[\cos^{-1}(-\bar{z})]} \quad (9.4.19)$$

This downwash distribution is shown in Fig. 9.4.6 for a typical unswept tapered wing. The downwash a few chord length or more downstream from a tapered wing, as obtained by using Eq. (9.4.10) in Eq. (9.2.44), is

$$V_y(\bar{x}, \bar{y}, \bar{z}) = -\frac{V_\infty C_L \kappa_v}{\pi^2 R_A} \left\{ \frac{\kappa_b - \bar{z}}{\bar{y}^2 + (\kappa_b - \bar{z})^2} \left[ 1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b - \bar{z})^2}} \right] + \frac{\bar{x}}{\bar{x}^2 + \bar{y}^2} \left[ \frac{\kappa_b - \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b - \bar{z})^2}} + \frac{\kappa_b + \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b + \bar{z})^2}} \right] + \frac{\kappa_b + \bar{z}}{\bar{y}^2 + (\kappa_b + \bar{z})^2} \left[ 1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b + \bar{z})^2}} \right] \right\} \quad (9.4.20)$$

$$\kappa_v = 1 + \sum_{n=2}^N \frac{b_n}{b_1} \sin(n\pi/2) \quad (9.4.21)$$

$$\kappa_b = \frac{1}{\kappa_v} \left[ \frac{\pi}{4} + \sum_{n=2}^N \frac{n b_n}{(n^2 - 1) b_1} \cos(n\pi/2) \right] \quad (9.4.22)$$

Numerical results for  $\kappa_v$  and  $\kappa_b$  are plotted in Figs. 9.4.7 and 9.4.8.

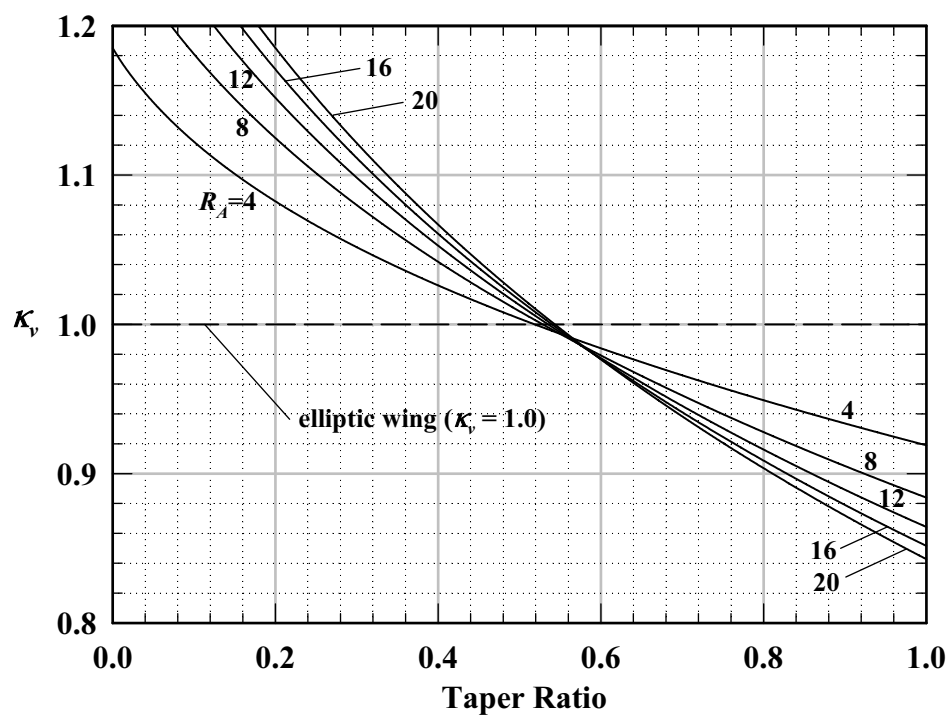


Figure 9.4.7. The wingtip vortex strength factor from Prandtl's lifting-line theory.

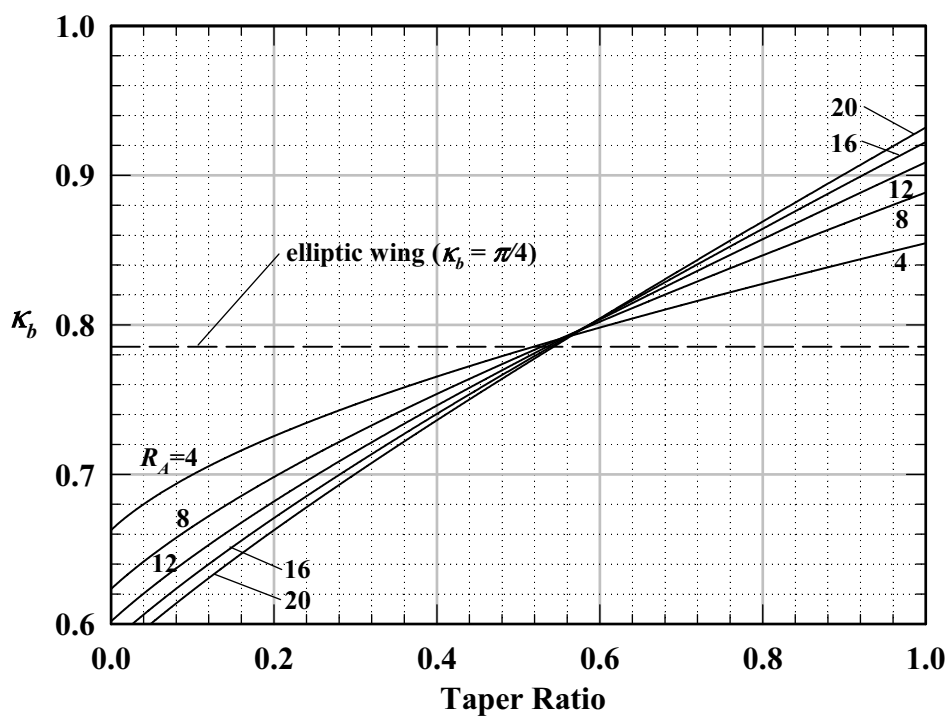


Figure 9.4.8. The wingtip vortex span factor from Prandtl's lifting-line theory.

**Example 9.4.1.** Consider a rectangular wing of aspect ratio 8 having no sweep, no dihedral, no geometric twist, and no aerodynamic twist. The wing is constructed with a thin symmetric airfoil section. For an angle of attack of  $5^\circ$ , estimate the lift coefficient for the wing, the induced drag coefficient for the wing, and the downwash angle induced by the wing, four chord lengths downstream from the center of the wing quarter-chord.

**Solution.** From Figs. 9.4.4, 9.4.5, 9.4.7, and 9.4.8 (or using the program *taper*, which may be downloaded from the web site at “[www.aero.mae.usu.edu](http://www.aero.mae.usu.edu).”), for an aspect ratio of 8.0 and a taper ratio of 1.0, the following results are obtained,

$$\kappa_L = 0.039$$

$$\kappa_D = 0.068$$

$$\kappa_v = 0.884$$

$$\kappa_b = 0.888$$

Using a airfoil section lift slope of  $2\pi$  and a zero lift angle of attack of 0.0 for the thin symmetric airfoil section, Eqs. (9.4.13) and (9.4.14) give

$$C_L = \frac{\tilde{C}_{L,\alpha}(\alpha - \alpha_{L0})}{(1 + \tilde{C}_{L,\alpha}/\pi R_A)(1 + \kappa_L)} = \frac{2\pi(5\pi/180 - 0.0)}{(1 + 2\pi/8\pi)(1.039)} = \frac{0.5483}{(1.25)(1.039)} = 0.422$$

Eq. (9.4.16) gives

$$C_{Di} = \frac{C_L^2}{\pi R_A}(1 + \kappa_D) = \frac{(0.422)^2}{8\pi} 1.068 = 0.00757$$

Four chord lengths behind the center of the wing quarter-chord, we have

$$\bar{x} = \frac{x}{b/2} = \frac{4\bar{c}}{b/2} = \frac{8\bar{c}b}{b^2} = \frac{8S}{b^2} = \frac{8}{R_A} = 1$$

$$\bar{y} = \frac{y}{b/2} = 0$$

$$\bar{z} = \frac{z}{b/2} = 0$$

and from Eq. (9.4.20) the downwash angle is

$$\begin{aligned} \varepsilon_d &= \frac{-V_y(1,0,0)}{V_\infty} = \frac{2C_L \kappa_v}{\pi^2 R_A} \left\{ \frac{1}{\kappa_b} \left[ 1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + (\kappa_b)^2}} \right] + \frac{\kappa_b}{\bar{x}\sqrt{\bar{x}^2 + (\kappa_b)^2}} \right\} \\ &= \frac{2(0.422)(0.884)}{8\pi^2} \left\{ \frac{1}{0.888} \left[ 1 + \frac{1}{\sqrt{1 + (0.888)^2}} \right] + \frac{0.888}{1\sqrt{1 + (0.888)^2}} \right\} = 0.02487 \text{ rad} = 1.425^\circ \end{aligned}$$

In **summary** the finite tapered wing with no sweep, no dihedral, no geometric twist, and no aerodynamic twist can be designed such that the induced drag is close to the optimum value of an elliptic wing. The chord length for this wing varies linearly with the spanwise coordinate and the circulation distribution can be written in terms of a Fourier series with coefficients,  $b_n$ , that are independent of angle of attack. Numerical results can be obtained from Figs. 9.4.4, 9.4.5, 9.4.7, and 9.4.8 or from the program *taper*, which may be downloaded from the web site at “[www.aero.mae.usu.edu](http://www.aero.mae.usu.edu).”

The **circulation distribution** for a tapered wing is

$$\Gamma(\theta) = 2bV_\infty(\alpha - \alpha_{L0}) \sum_{n=1}^N b_n \sin(n\theta); \quad \theta = \cos^{-1}(-2z/b) \quad (9.4.11)$$

$$\sum_{n=1}^N b_n \sin(n\theta) \left[ \frac{2R_A(1+R_T)}{\tilde{C}_{L,\alpha}[1-(1-R_T)|\cos(\theta)|]} + \frac{n}{\sin(\theta)} \right] = 1 \quad (9.4.9)$$

We can write the **lift and induced drag coefficients** as

$$C_L = C_{L,\alpha}(\alpha - \alpha_{L0}), \quad C_{L,\alpha} = \frac{\tilde{C}_{L,\alpha}}{(1 + \tilde{C}_{L,\alpha}/\pi R_A)(1 + \kappa_L)}, \quad \kappa_L = \frac{1 - (1 + \pi R_A/\tilde{C}_{L,\alpha})b_l}{(1 + \pi R_A/\tilde{C}_{L,\alpha})b_l} \quad (9.4.13)$$

$$C_{Di} = \frac{C_L^2}{\pi R_A e_s}; \quad e_s = \frac{1}{1 + \kappa_D}; \quad \kappa_D = \sum_{n=2}^N n \left( \frac{b_n}{b_l} \right)^2 \quad (9.4.16)$$

The **downwash induced near a tapered wing** is,

$$V_y(\bar{x}, \bar{y}, \bar{z}) = -\frac{V_\infty C_L}{\pi R_A} \sum_{n=1}^N \frac{b_n}{b_l} \frac{1}{\pi} \int_{\phi=0}^{\pi} \left\{ \frac{n \cos(n\phi)[\bar{z} + \cos(\phi)]}{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2} + \frac{n\bar{x} \cos(n\phi)[\bar{z} + \cos(\phi)]}{\{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2\} \{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{1/2}} + \frac{\bar{x} \sin(n\phi) \sin(\phi)}{\{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{3/2}} \right\} d\phi \quad (9.4.18)$$

The **downwash a few chord lengths or more downstream from a tapered wing** is

$$V_y(\bar{x}, \bar{y}, \bar{z}) = -\frac{V_\infty C_L \kappa_v}{\pi^2 R_A} \left\{ \frac{\kappa_b - \bar{z}}{\bar{y}^2 + (\kappa_b - \bar{z})^2} \left[ 1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b - \bar{z})^2}} \right] + \frac{\bar{x}}{\bar{x}^2 + \bar{y}^2} \left[ \frac{\kappa_b - \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b - \bar{z})^2}} + \frac{\kappa_b + \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b + \bar{z})^2}} \right] + \frac{\kappa_b + \bar{z}}{\bar{y}^2 + (\kappa_b + \bar{z})^2} \left[ 1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b + \bar{z})^2}} \right] \right\} \quad (9.4.20)$$

$$\kappa_v = 1 + \sum_{n=2}^N \frac{b_n}{b_l} \sin(n\pi/2), \quad \kappa_b = \frac{1}{\kappa_v} \left[ \frac{\pi}{4} + \sum_{n=2}^N \frac{nb_n}{(n^2 - 1)b_l} \cos(n\pi/2) \right]$$